

CSE251

Assignment 1

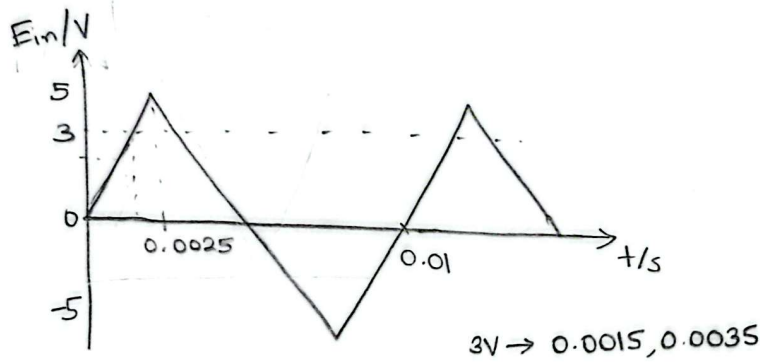
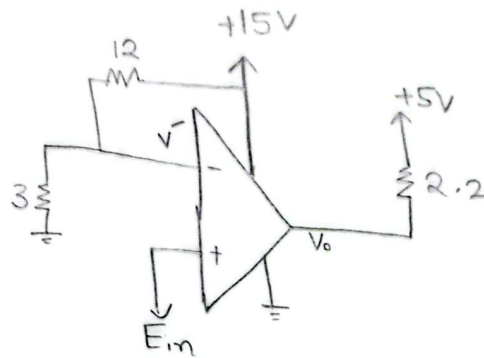
Section - 7B

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24301500

2.1

10.



$$\frac{3}{3} = \frac{V^- - 15}{12}$$

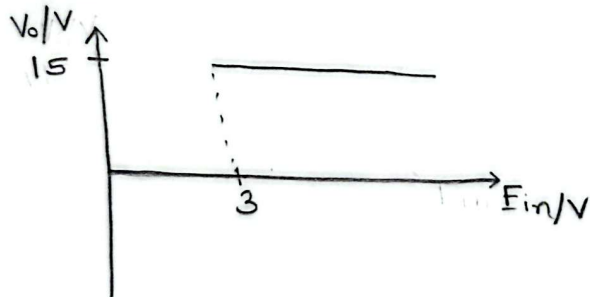
$$\frac{0 - V^-}{3} = \frac{V^- - 15}{12}$$

$$-\frac{V^-}{3} = \frac{V^-}{12} - \frac{15}{12}$$

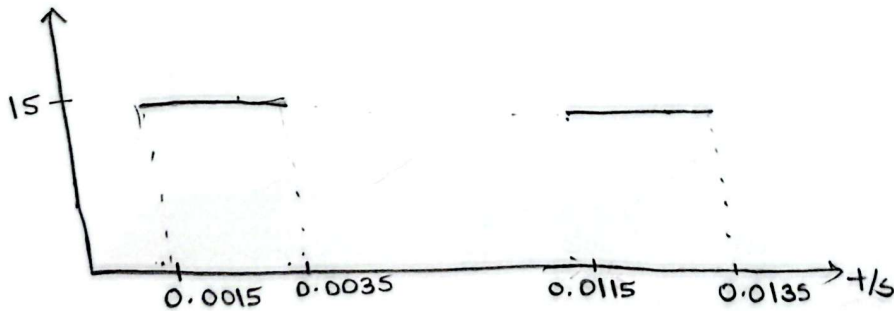
$$-\frac{5}{12} V^- = -\frac{15}{12}$$

$$V^- = 3V$$

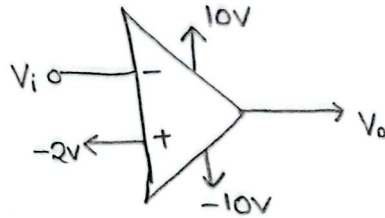
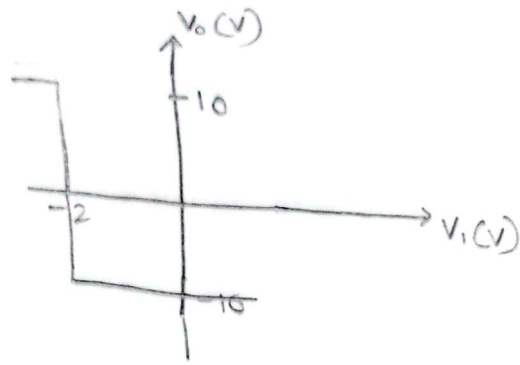
a) V_o vs E_{in}



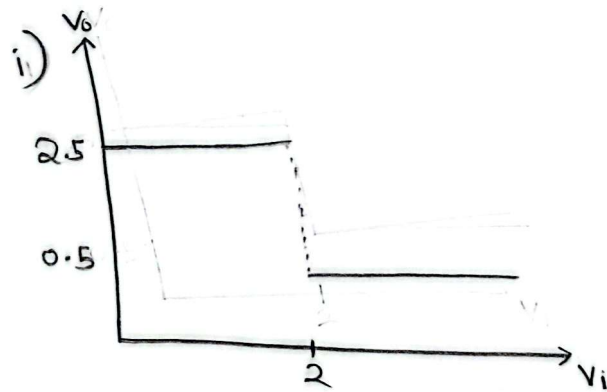
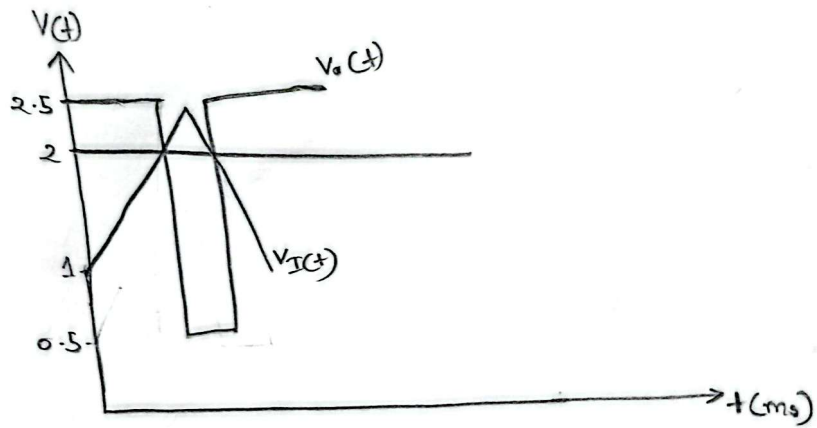
b) V_o vs t



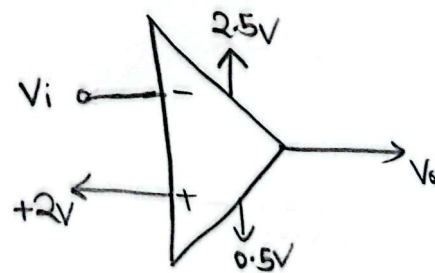
12.



14.



ii)

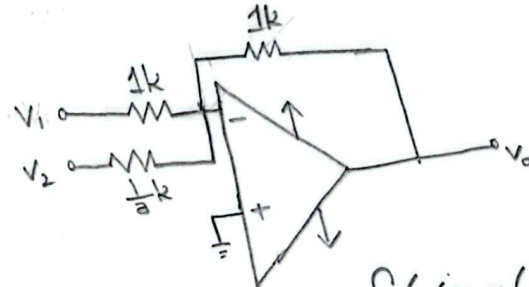


2.2

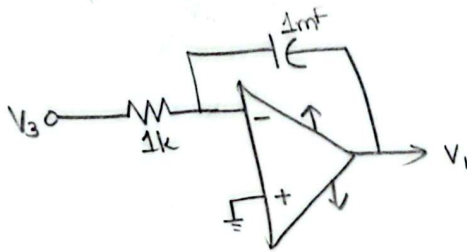
1(VI).

$$f = \int (6x - \frac{3d^2y}{dt^2}) dt - 3 \frac{dz}{dt}$$

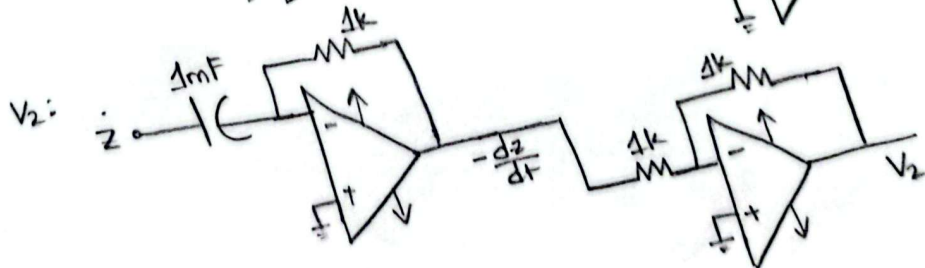
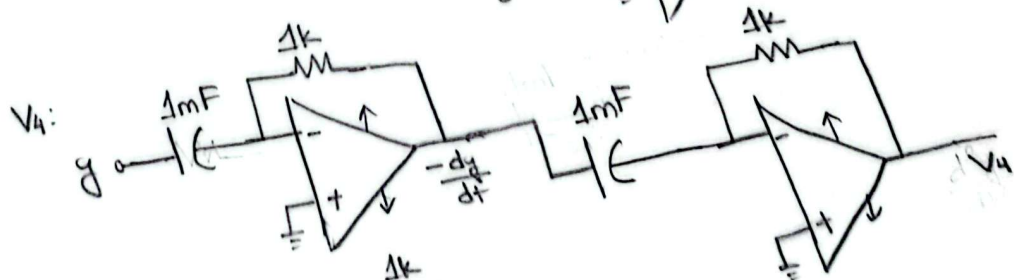
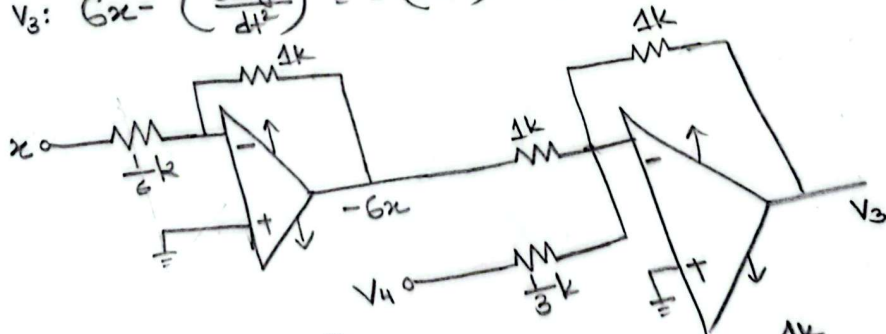
$$= - \int \underbrace{(\frac{3d^2y}{dt^2})}_{V_1} - 6x \underbrace{dt}_{V_2} - 3 \underbrace{\frac{dz}{dt}}_{V_3}$$



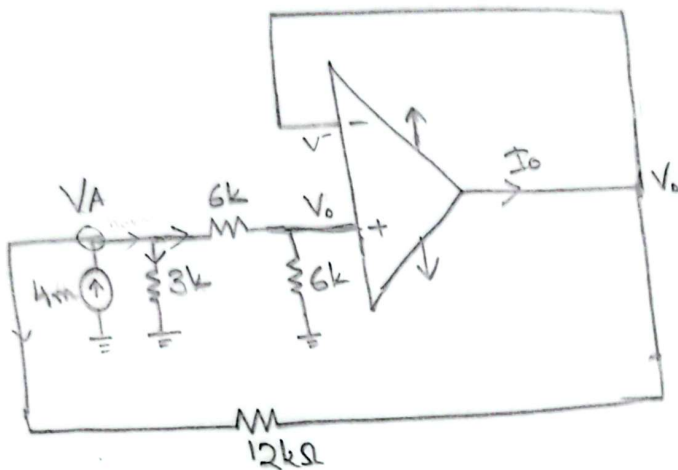
$$V_1: \int (\frac{3d^2y}{dt^2}) - 6x dt = - \int \underbrace{(6x - (\frac{3d^2y}{dt^2}))}_{V_3} dt$$



$$V_3: 6x - (\frac{3d^2y}{dt^2}) = -(-6x) - (\frac{3d^2y}{dt^2}) = -(-6x) - 3(\frac{d^2y}{dt^2})$$



13. V_o and I_o



KCL at V_A

$$4\text{mA} = \frac{V_A - V_o}{12k} + \frac{V_A - 0}{3k} + \frac{V_A - 0}{12k}$$

$$4 = \frac{2V_o - V_o}{12} + \frac{2V_o}{3} + \frac{2V_o}{12}$$

$$4 = \frac{1}{12}V_o + \frac{2}{3}V_o + \frac{2}{12}V_o$$

$$4 = \frac{11}{12}V_o$$

$$V_o = 4.36\text{V (Ans)}$$

$$\frac{V_o - V_A}{12} = I_o \quad [A_s \text{ no current flow from } V^-]$$

$$\frac{4.36 - (2 \times 4.36)}{12} = I_o$$

$$I_o = -0.363\text{mA (Ans)}$$

$$V_o = A(V^+ - V^-) \quad \text{but } V^- = V_o$$

$$V_o = A(V^+ - V_o)$$

$$V_o = A(V^+) - AV_o$$

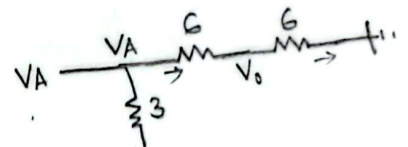
$$(1+A)V_o = AV^+$$

$$V_o = \frac{AV^+}{1+A}$$

$$V_o = \frac{V^+}{\frac{1}{A} + 1} \Rightarrow \frac{1}{A} \approx 0 \text{ as } A \gg 1$$

$$V_o = V^+$$

no current enters through V^+

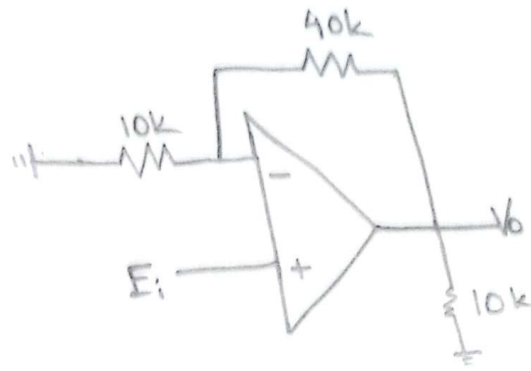


$$\frac{V_A - V_o}{6} = \frac{V_o - 0}{6}$$

$$\frac{V_A - V_o}{6} = \frac{V_o}{6}$$

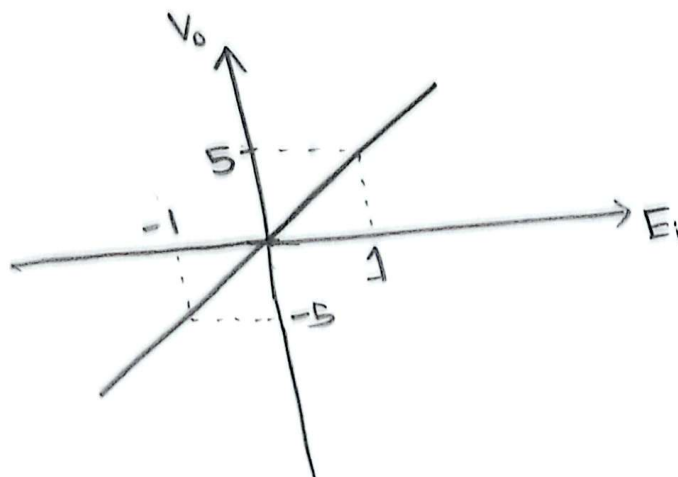
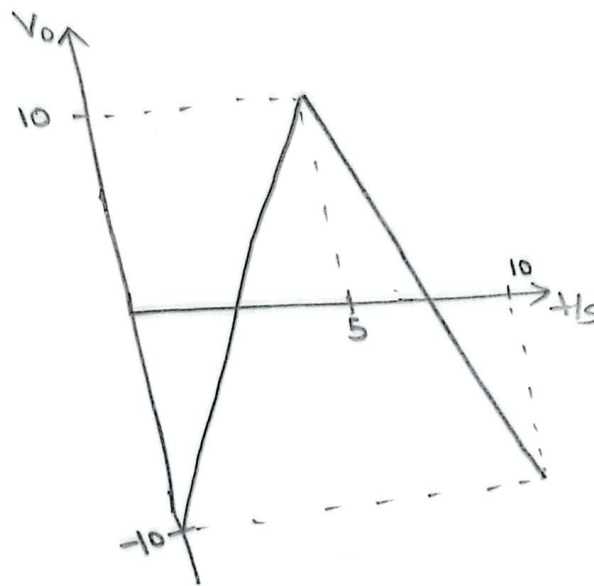
$$V_A = 2V_o$$

24.



$$A = \left(1 + \frac{40}{10}\right)$$

$$= 5$$



26ii)

